ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** A is true but R is false.	A is true; R is false because animal cells lack a cell wall.
2	**(b)** 2 mm	2 x 1000 = 2000 micrometres = 2 mm.
3	**(b)** stops functioning properly	Nucleus controls the cell.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Chloroplasts carry out photosynthesis, which is why plants (which make their own food) have them.
5	**(c)** 0.05 mm	5 / 100 = 0.05 mm.
6	**(c)** yeast is more active at warm temperature	Warmth boosts fermentation.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** makes food by photosynthesis	Chloroplasts trap light for photosynthesis.
9	**(b)** tissues	Tissue means many cells.
10	**(a)** Both A and R are true and R is the correct explanation of A.	The space allows more haemoglobin.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Actual = 5 / 500 = 0.01 mm (1); = 10 micrometres (1).	1+1
12	Amoeba has only a flexible membrane (1); Plant cell has a rigid wall (1).	1+1
13	All organisms are built of cells (1); Cells combine into tissues, organs and bodies (1).	1+1
14	Cork consists of dead cells (1); Only the walls remain after the contents are lost (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Onion cells are rectangular with a cell wall; cheek cells are rounded or irregular without a wall (1); Both show a nucleus and cytoplasm; the onion cell has a large vacuole (1); Stains such as safranin or methylene blue make the parts visible (1).	1+1+1
16	Digestion: food vacuole (1); Excretion: contractile vacuole (1); Movement: pseudopodia (1).	1+1+1
17	Antibiotics act on bacteria (1); Fungi have different cell walls (1); Fungicides are needed (1).	1+1+1
	OR: Virus is smaller than bacterium (1); Virus is non-cellular; bacterium is a cell (1); Bacterial diseases need antibiotics; viral diseases need vaccines/antivirals (1).	1+1+1
18	No division of labour (1); Complex functions cannot be done (1); Organisms would stay simple (1).	1+1+1
	OR: Egg cell is about 0.1 mm (1); Bacterium is about a micrometre (1); A cell's size depends on its function (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Onion peel stained with safranin (1); Cheek cells stained with methylene blue (1); Plant: rectangular, wall, large vacuole (1); Animal: rounded, irregular, no wall (1); Both: nucleus and cytoplasm (1).	1+1+1+1+1
	OR: Cells in a multicellular body divide the work among themselves (1); Nerve cells are long and branched to carry messages (1); Red blood cells carry oxygen (1); Root hair cells have long projections to absorb water (1); Specialisation makes the body work efficiently (1).	1+1+1+1+1
20	Objective = 800 / 10 = 80x (1); Actual = 4 / 800 mm (1); = 0.005 mm (1); = 5 micrometres (1); This size suggests a bacterium (1).	1+1+1+1+1
	OR: Milk is warmed to lukewarm (1); A spoon of curd containing Lactobacillus is added (1); Bacteria multiply and change lactose to lactic acid (1); Milk sets as curd (1); Useful for digestion (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option

attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** virus, bacterium, red blood cell	Smaller to larger.
2	**(d)** Flu is caused by a virus	Viruses are not affected.
3	**(d)** A is false but R is true.	A is false: the cells were dead. R is true.
4	**(c)** specialised for movement	Tail helps in swimming.
5	**(c)** it has a rigid cell wall	Wall fixes the shape.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Reason explains it.
7	**(b)** cell size varies widely among organisms	Sizes range from micrometres to centimetres.
8	**(d)** wall is outside the membrane	Wall is outermost.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The nuclear membrane is absent.
10	**(b)** Lactobacillus bacteria	Curd bacteria.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Pathogen: disease-causing microbe, e.g. typhoid bacterium (1); Decomposer: recycles dead matter, e.g. soil fungi (1).	1+1
12	Many are harmless or helpful (1); Some make vitamin K and help in digestion (1).	1+1
13	Antibiotics act on bacterial cell structures (1); Viruses have no cells and multiply only inside host cells (1).	1+1
14	Yeast ferments sugar (1); Releases carbon dioxide that fills the balloon (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Nucleus: manager (1); Cytoplasm: working floor (1); Membrane: gate (1).	1+1+1
16	They have different functions (1); Different parts become active (1); Cells specialise during development (1).	1+1+1
17	Stain not used or not enough (1); Cells are folded (1); Wrong focus (1).	1+1+1
	OR: Dead cells lose their living contents (1); Walls remain to give support (1); Such cells occur in xylem and cork (1).	1+1+1
18	Unicellular divide into two (1); Multicellular produce gametes/seeds (1); Multicellular are more complex (1).	1+1+1
	OR: Large surface area compared to volume (1); Materials move in and out quickly (1); Cell activities are efficient (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Benefits: curd, bread, antibiotics (1); Decay and recycling (1); Harms: diseases, spoilage (1); Hygiene prevents spread (1); Safe food storage and boiling water (1).	1+1+1+1+1
	OR: Cell cannot control its functions or divide (1); Genetic information is lost (1); It will die (1); RBC lacks nucleus but works for some time (1); Because it has a fixed job only (1).	1+1+1+1+1
20	Nerve: long, branched (1); RBC: disc, no nucleus (1); Root hair: long projection (1); Muscle: spindle, contract (1); Guard cell: bean-shaped, control pores (1).	1+1+1+1+1
	OR: Wall: support (1); Membrane: control (1); Vacuole: storage (1); Wall is porous (1); Membrane is selective (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** bigger organisms have more cells, not bigger cells	Body size depends on the number of cells.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Shape suits function.
3	**(a)** virus	Viruses are too small.
4	**(d)** lacks chlorophyll and cannot make food	Fungi are heterotrophic.
5	**(c)** A is true but R is false.	A is true; R is false.
6	**(c)** organ it belongs to	Organs depend on tissues.
7	**(c)** cause disease in host organisms	Disease-causing.
8	**(c)** A is true but R is false.	A is true; R is false as fungi lack chlorophyll.
9	**(a)** different organs doing different jobs in a body	Typical of multicellular organisms.
10	**(d)** A tissue is made of cells of the same kind.	Tissue is similar cells.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Organelle: part inside a cell (1); Organ: part of a body made of tissues (1).	1+1
12	Wall is permeable; does not control materials (1); Membrane regulates entry and exit (1).	1+1
13	Stain brings out the cell parts (1); Cover slip keeps the specimen flat and protected (1).	1+1
14	Leaves get light (1); They are the site of photosynthesis (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	20 micrometres = 0.02 mm (1); 0.07 / 0.02 (1); = 3.5, about 3 to 4 cells (1).	1+1+1
16	No wall: draw outermost wall (1); Add a large central vacuole (1); Also label chloroplasts and nucleus (1).	1+1+1
17	Do not touch the lens (1); Handle slides gently (1); Do not use direct sunlight (1).	1+1+1
	OR: Number of doublings = 120 / 20 = 6 (1); 2^6 (1); = 64 bacteria (1).	1+1+1
18	Place a drop of curd water or pond water (1); Stain with methylene blue (1); Use the highest power (1).	1+1+1
	OR: Showed that germs cause disease (1); Led to cleanliness, boiling and sterilisation (1); Vaccines were developed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cells: nerve cell, red blood cell (1 mark for two examples) (1); Tissues: muscle tissue, nervous tissue (1); Organs: heart, lungs (1); Organ systems: digestive, circulatory (1); Organism: the human body made of many cells working together (1).	1+1+1+1+1
	OR: Cell: photosynthetic cell (1); Tissue: palisade tissue (1); Organ: leaf (1); System: shoot system (1); Organism: whole plant (1).	1+1+1+1+1
20	Bacteria: no nucleus; typhoid (1); Fungi: wall, no chlorophyll; ringworm (1); Protozoa: with nucleus; malaria (1); Size ranges differ (1); All can be seen under a microscope (1).	1+1+1+1+1
	OR: Leeuwenhoek: single lens (1); Hooke: compound microscope (1); Greater magnification followed (1); Light microscope limit: cannot see viruses (1); Electron microscopes see viruses (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.